Agentic AI for Personalized Career Guidance: Concepts, Architecture, and Research Directions

Rahul Vijaykumar Nalage*
Research Scholar
M Tech –ISA
MIT ADT
Pune India
rahul.nalage@gmail.com

Reena Gunjan
Program Head
M Tech –ISA
MIT ADT
Pune India
reena.gunjan@mituniversity.edu.in

Corresponding author: Rahul Vijaykumar Nalage (rahul.nalage@gmail.com)

Abstract— **This paper contains a detailed description of Agentic (artificial intelligence) AI) and its possible implementation in the personal career guidance system. The changes occurring in the labor markets along the digital spectrum necessitate the new, adaptive, context-sensitive development models that go beyond the stagnant recommender systems. The agentic AI that is typified by autonomy, planning, memory, and reflective learning can provide new career systems avenues that continuously adapt the individual desire to changing market environments. This review brings together recent conceptualizations and taxonomies of agentic AI and suggests our four-layer conceptual model, Perception, Cognition, Action, and Reflection. This framework recognizes how agentic capabilities can be used to facilitate dynamic user modeling, proactive guidance activities and re-integrating iterative feedback. Our raised points imply on explicability, fairness, and ethical decision-making in high stakes. We also find that there are several significant research issues, which bring to the fore the necessity of scalable integration, knowledge representation programs using knowledge graph, the development of accountability protocols and, validated evaluation using synthetic simulations. Lastly, we outline a research agenda that puts forward interdisciplinary methods to consider various parties involved in governance, and to establish agentic career guidance systems that are transparent, explainable and trustworthy.**

Keywords—***Agentic AI, Career Guidance, Personalization, Knowledge Graphs, Explainability, Ethics, Autonomous Systems, Recommender Systems, Human–AI Collaboration, Educational Data Mining, Labor Market Analytics, Career Path Recommendation, Reflective Learning.***

I. Introduction

The employment market is experiencing an unprecedented change that is driven by the accelerated developments in the background of automation, artificial intelligence (AI) and digital technologies.

This has put pressure on the need to have career guidance systems that have the ability to be versatile and context sensitive and at the same time constantly remind the user of his or her long term objectives in life and the dynamic nature of the labor market. The conventional recommender systems can offer suggestions concerning educational or professional courses, yet they are often extremely flawed.

As a rule, recommender systems are reactive, non-proactive, have no ability of an agent to be long-term and proactive, and constantly update and revise plans and strategies depending on user and market conditions.

An agentic artificial intelligence (Agentic AI) refers to a type of artificial intelligence that is capable of exhibiting autonomy, planning, episodic memory and reflective learning, and represents an effective conceptual way of overcoming the limitations of the traditional AI agent, who largely limits their functions to current examples or tasks when they act under the supervision of a human agent.

The agentic systems are also expected to be more autonomous, dynamic and accountable, in return, supporting very individualized and dynamic careers amongst the users.

This contribution discusses the definition of Agentic AI against personalised career guidance by conducting a conceptual review that synthesizes new definitions and taxonomies of agentic systems into four conceptual levels, and discussing the ethical and explainability concerns, and governance issues.

© The Author(s) 2025
T. Ali et al. (eds.), *Proceedings of the 1st Engineering Data Analytics and Management Conference (EAMCON 2025)*, Advances in Engineering Research 289,
https://doi.org/10.2991/978-94-6463-978-0_50



Fig. 1. Agentic Artificial Intelligence for Personalized Career Guidance

The article finishes with a conclusion and a future agenda that includes some of the expected issues in research, but it should be emphasized that the goal is to allow agentic AI to positively affect users and enhance their empowerment and trust as well as have a successful career.

II. Related Work

Career advisory and recommendation systems have been growingly incorporating aspects of artificial intelligence (AI), machine learning (ML), and deep learning (DL) techniques to tackle the challenges of aligning personal competencies, preferences and ambitions with different opportunities in the work force.

A number of researches have been done on career mobility and labor market modeling. A single study by Zha et al. [1] introduced an uncertainty-conscious graph autoencoder model to predict the potentially desirable occupational shift, which gives some understanding of the movement of the labor market in a more macro context.

There was another study conducted by Liu and Ge [3], which used a joint deep learning model to learn job and employee embeddings on platforms like LinkedIn and Glassdoor, and they combined the embeddings to aid in better suggestions on job titles based on the similarity perspective.

In the meantime, an equivalent field of research has assessed student recommendation systems based on personalized career paths. Indicatively, Siswipraptini et al [2] suggested a personalized career-path recommendation model based on Naive Bayes and tested among IT students in Indonesia.

Qamhieh et al. [5] created a recommender system (PCRS) which is a fuzzy-logic-based system tailored to engineering students in Palestine. The studies identified the role of personality data and academic data in career advice in both studies.

Career re-education and re-skilling by means of AI integration are also the subject of research. Ashrafi et al. [4] introduced a fast re-education model grounded on resume data and trained neural networks using CNN to avoid labor-market-disruption, which was caused by automation as well as by COVID-19 caused disruptions.

Raja and Priya [6] developed a profound earning-based model that utilizes Natural Language Processing, Recurrent Neural Networks, and CNNs to assess life skills and give a career suggestion and stressed the importance of explainable AI in the decision-making process.

A number of recent articles have discussed the emergence of AI-based career counseling app. Waikar and Tamhane [11] used generative AI to enhance the career counseling process among students whereas Barbadekar and Bhagwan [13] generated a kind of career guide which combined the data of prediction provided by the ML with a chatbot interface to maintain an interactive session.

Jawhar and Bitar [14] introduced an AI-assisted web-based platform to support customized decision-making regarding the university and career outcomes, and Bhargava and Koul [18] advanced their methods by simplifying career advice using a multilingual generative AI named CounselAI.

Hybrid architectures based on deep learning models have also been demonstrated to be effective. Kumar and Garg [10] developed a model using CNN-LSTM architecture that is oriented toward predictive career development advice and entrepreneurial development, which achieved higher performance on the task as compared with a generic ML using career guidance application use. These publications indicate that there is a possibility of the usefulness of the adaptability of deep learning strategies in the space and time data representation of career prediction problems.

New research has started to use agentic and conversational AI methods. Wang et al. [16] suggested the AdaptJobRec AdaptJobRec is an agentic career recommendation system based on LLM that has the potential to balance the accuracy of reasoning with the latency of response.

Similarly, Ramesh and Hema [7] proposed a career guidance system with AI-based SavvyAI, which is based on predictive analytics and user-focused design. These works are a change to more adaptive, context sensitive and interactive systems.

Moreover, studies have been put on career guidance in higher education as well as youth services. According to Fawaid and Baharun [9], AI-based decision support can improve career management in the universities.

Digital media literacy study was suggested by Petkov [17] as a way of assisting in modernization of career services, including the youth career services. The NEFELE project [15] research topics also cited approaches of training teachers in career support as superior ways of attaining career guidance as opposed to considering career path planning and development of students in a comprehensive manner in both methods of using AI-based student support.

Overall, the study of career guidance and career management using AI-based systems has moved ahead in many aspects but most systems remain constrained in terms of scaling and interpretations or proactive to the longer-term career planning and development.

This leads to the rationale behind investigating college and agentic AI models as a component of broader strategies to career guidance systems that examine autonomous career guidance, planning and reflective learning and reactive dispositions to consideration of alternative careers an adaptive career guidance solution.

III. AGENTIC AI CONCEPTUAL FRAMEWORK

We propose a four-layer conceptual framework for integrating Agentic AI into personalized career guidance systems. The model is designed to capture user-specific contexts, enable autonomous decision-making, and provide adaptive career interventions over time.

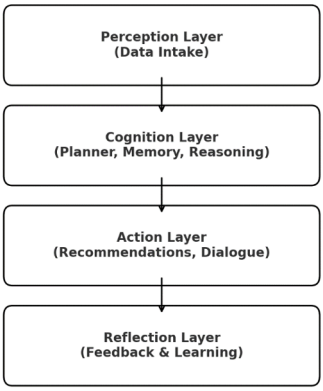


Fig. 2. Agentic AI Conceptual Framework Layers

1. Perception Layer

This layer is in charge of the collection and preprocessing of heterogeneous input signals. These are user-specific information like academic history, interests, psychometric tests, and contact histories, and external labor market information such as job advertisements, professional standards, and skills taxonomies.

Combining both personal and market-level data, the Perception Layer creates a wholesome picture of the career landscape.

2. CognitionLayer

The central part of the framework is Cognition Layer that stores persistent memory structures and dynamic goal models. It facilitates long-term and short-term planning by breaking down user goals into sub goals which can be achieved.

The Cognition Layer has language models (LLMs) that are domain-centered transformer-based models. Career plot of knowledge in such a way that it enables one to think logically.

Reasoning mechanisms are also implemented in this layer, which uses knowledge graphs and domain ontologies to harmonize the aspirations of users with the opportunities that emerge in the market.

3. Action Layer

According to cognitive outputs, the Action Layer produces individual interventions.
These are career advice, planned learning, and dialogue-based guidance provided via conversational platforms.

The layer makes sure that interventions are contextually relevant and in line with user preference and changing market conditions.

4. Reflection Layer

To pursue flexibility and self-development, the Reflection Layer considers the usefulness of previous interventions through the introduction of user feedback and outcomes.

The feedback loop allows one to make continuous updates to the memory, recalibrate goals, and optimize system policy. The framework encourages long-term flexibility and resiliency to evolving trends in the labor market by instilling the element of reflective learning.

Reinforcement learning of the Human Feedback (RLHF) and the schemes of preference optimization are now implemented to enlarge the recommendation pathways with the passage of time.

The feedback channels which will be included in the Reflection Layer are definite scores on relevance of the recommendations, unconscious signs of behavior (clicks, time of attention, watched recommendations),and way of natural language responses through chat. These signals are transformed into reward scores and they are optimized to user preference vectors.

The whole architecture, as shown in Fig. 2, points out the interaction between these four layers as well as the importance of the reflection and adaptation in maintaining a successful and personalized career guidance.

IV. ETHICS AND EXPLAINABILITY

The use of agentic AI systems in career guidance poses special ethical issues because of its independence and long-term effects on the professional path of its users.

The risks that are possible are objective drift, the system goals do not necessarily reflect what the user intends; inappropriate autonomy, when agents make decisions that are outside acceptable limits; and opaque decision making, making it difficult to trust a decision maker when recommendations have a direct impact on the employability and livelihood.

As a solution to these concerns explainability should be made a first-class capability. Interpretable reasoning should be delivered by agentic systems, showing the hierarchies of goals, the bases of recommendation that are evident, and revealing representations of uncertainty. This explainability will make sure that users and other stakeholders do not just know what is being recommended but also why it is suitable.

Moreover, accountability and ethical adherence should be upheld by governance systems. These are audit trails on the decision paths, human-in-loop points to sensitive interventions, and clear user consent models, which allow individuals to have the remaining power to control their data and data.

Lastly, evaluation systems must not only measure accuracy in prediction, but also include measures of fairness, transparency and user satisfaction.

When integrated into validation pipelines, agentic AI systems can support ethical principles as well as user expectations, which will facilitate more trust, adoption, and long-lasting efficacy of career guidance systems.

V. Knowledge Graphs and Data Modelling

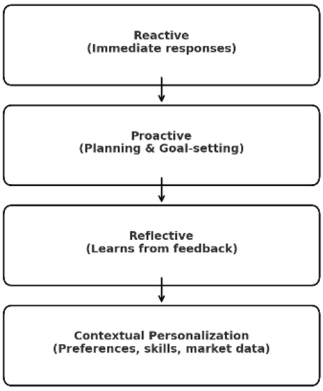


Fig. 3. Agentic AI taxonomy for personalization behavior.

The knowledge graphs (KGs) represent structured and interpretable representations that can be used to interconnect the skills, occupations, learning materials, and the assessment results. KGs can be used to encode relations between these entities, and thus, can be used to perform machine reasoning, such as to derive prerequisite skills or transferable domain-cutting-edge competencies.

This ability not only increases personalization but also explainability since path-based justifications can follow the trail of reasoning behind system recommendations.

The teachers present the information about the currency of the curriculum and ability testing. The employers offer employment, models of skills and recruitment expectations.
Policymakers are the ones who introduce ethical standards, privacy and fairness compliance standards. Audit of the outputs of recommendation is also done by an oversight committee on a regular basis in order to provide fairness and transparency.

In the suggested architecture, the Cognition Layer uses KGs to get contextual knowledge that facilitates planning and decomposing long-term goals. Recent studies have promptly shown the usefulness of dynamically built job-skill knowledge graphs built on by sources like O*NET and other labor-market cues.

These papers emphasize the ability of large-scale labor data to be converted into structured ontologies in order to make career-path modeling effective.

One of the engineering issues is related to the incorporation of KG updates into agent memory and planner modules. The system should be dynamic to align the knowledge updates with the user-specific pathways, and hence the recommendations up to date, reliable, and personalized because labor markets are constantly changing.

It is through this synergy between KG based reasoning and reflective memory mechanisms that agentic AI can be developed in career guidance.

As shown in Fig. 3, the taxonomy of Agentic AI personalization behaviors, the role of KGs in supporting interpretability and adaptive interventions, is demonstrated.

Among the diluted examples that have been advanced to show the applicability of the suggested framework include that of an undergraduate student in her final year in the discipline of engineering and who requires some guidance on the career path that she must pursue. Data are used in collecting academic transcripts, skills tests, project portfolio data and preference inputs.

The Cognition Layer takes the reasoning on the knowledge-graphs to generate a list of career paths, e.g. data analyst and embedded engineer, rank and skill-gap analysis and learning pathways. Reflection Layer will also ensure that the successive suggestion would be developed upon user response like preference switch and rating of relevance following the provision of a user with a certain recommendation. The case study is an excellent illustration on how the Agentic Adaptation operates in natural settings and the pilot deployment studies are therefore convenient in the future.

VI. Research Challenges

Although these improvements have happened recently, there are still several important research issues in achieving agentic AI in career guidance:

Labor market real-time integration and data quality: To achieve an effective personalization, the labor market needs to be provided with high quality and real-time labor market signals. Some of the challenges are noise in job postings, variable taxonomies and heterogeneous data sources integration.

Ongoing learning and memory control: Agentic systems have to adapt plasticity (gathering new knowledge) and stability (long-term knowledge) without catastrophic forgetting in developing user models.

Ethics and governance: There is the risk of bias amplification, no avenue of redress to the user adversely affected by any recommendation, and excessive dependence on obscure autonomy. Systems of governance are required to bring about fairness, accountability and transparency.

Assessment: Career guidance needs longitudinal assessment as opposed to traditional recommendation activities. Virtual benchmarks, customer survey, and unbiased measures have to be established to evaluate results in totality.

Scalability: To serve populations in scale, and without compromising the quality of personalization, computational efficiency is needed in situations demanding global and multilingual support.

The system is capable of working on distributed vectors databases. We also have semantic retrieval that can be successfully implemented using (e.g., FAISS, Milvus) and federated learning. It may be trained through customization model devoid of sensitive user data centralization. Embedding Multilingualism Multilingual embedding (mBERT or XLM-R) are capable of supporting Multilingualism.

VII. Research Agenda

The challenges posed by agentic AI evolution on career guidance are documented below as proposed strategies for the next few years.

(a) Initial Taxonomic and Benchmarking Constraints. Defining agentic AI taxonomy and establishing career guidance benchmarking tasks will enable systematic and cohesive assessments to determine fluid systems that can be equilibrated and regularized.

(b) Knowledge Graphs and Privacy-Preserving Datasets. Engendering privacy-preserving knowledge graph schema will advance stakeholder trust and the collection and safeguarding of sensitive, career-related data that will be linked to open datasets.

(c) Explainability. Explainability and justification for career guidance recommendations will entail a specific goal hierarchy, the logic and uncertainty, as well as a proposed incremental achievement pathway within the norm.

(d) Longitudinal Simulations. Within a designated timeline to capture evolving user paths and systems dynamics, built-in and modeled simulation environments and validation tools will capture and demonstrate systematic adaptability, resilient, long-term effectiveness, and other outcomes as specified.

(e) Stakeholder Collaborative Governance. Jointly with educators, employers, and regulators, governance frameworks will be designed in a way that the systems remains harmonized with the prevailing social fabric, labor policy, and organizational systems.

VIII. Conclusion

The development of agentic AI suggests the possibility of adaptive, contextualized, proactive, goal-oriented, and reflective personalized career guidance systems. Unlike other recommendation systems, agentic systems autonomously plan and offer continuous support while exercising autonomy, self-evaluation, and recall.

The development of personalized guidance systems will require the design of AI frameworks, the representation of knowledge, ethical interdisciplinary governance, and holistic multi-disciplinary evaluation. In this paper, I drew from the emerging literature on ethical AI-powered career guidance and the broader AI career guidance literature, traversing the ethical governance and explainability challenges, and outline the initial contours of the literature on the kind of mentorship the discipline deserves, which remains largely unarticulated.

Integrating agentic reasoning with career data systems and the associated frameworks uncovers the need for the integration of evaluative fairness, transparency, and explainability. Ultimately, this will help AI deliver on its potential for equity and responsive career guidance. There is a need for field implementations that aim to and successfully demonstrate the effectiveness of such systems across diverse population parameters.

References

[1] R. Zha, C. Qin, L. Zhang, D. Shen, T. Xu, H. Zhu, and E. Chen, "Career mobility analysis with uncertainty-aware graph autoencoders: A job title transition perspective," IEEE Trans. Comput. Soc. Syst., vol. 11, no. 1, pp. 203–217, Feb. 2024.

[2] P. C. Siswipraptini, H. L. H. S. Warnars, A. Ramadhan, and W. Budiharto, "Personalized career-path recommendation model for information technology students in Indonesia," IEEE Access, vol. 12, pp. 41726–41738, Apr. 2024, doi: 10.1109/ACCESS.2024.3381032.

[3] H. Liu and Y. Ge, "Job and employee embeddings: A joint deep learning approach," IEEE Trans. Knowl. Data Eng., vol. 35, no. 7, pp. 6594–6607, Jul. 2023.

[4] S. Ashrafi, B. Majidi, E. Akhtarkavan, and S. H. R. Hajiagha, "Efficient resume-based re-education for career recommendation in rapidly evolving job markets," IEEE Access, vol. 11, pp. 108593–108605, Nov. 2023, doi: 10.1109/ACCESS.2023.3329576.

[5] M. Qamhieh, H. Sammanah, and M. N. Demaidi, "PCRS: Personalized career-path recommender system for engineering students," IEEE Access, vol. 8, pp. 214531–214542, Dec. 2020, doi: 10.1109/ACCESS.2020.3040338.

[6] R. Raja and P. M., "Optimizing career pathways: A deep learning-based life skills and recommendation framework," in Proc. 5th Int. Conf. Intell. Technol. (CONIT), Karnataka, India, Jun. 2025.

[7] V. Ramesh and H. M. S., "SavvyAI: AI enhanced personalized career guidance system," in Proc. 5th Int. Conf. Intell. Technol. (CONIT), Karnataka, India, Jun. 2025.

[8] M. Jawhar and M. Almotairi, "AI-powered career matching system for university students: Bridging education and employment," in Proc. IEEE/ACIS 23rd Int. Conf. Softw. Eng. Res., Manage. Appl. (SERA), Montreal, Canada, Jun. 2025, doi: 10.1109/SERA65747.2025.11154544.

[9] A. Fawaid and H. Baharun, "AI-based career management to improve the quality of decision making in higher education," in Proc. IEEE Integrated STEM Educ. Conf. (ISEC), New Jersey, USA, Mar. 2025, doi: 10.1109/ISEC64801.2025.11147274.

C. N. S. Kumar and S. Garg, "Hybrid CNN-LSTM model for predictive career guidance and entrepreneurial development in higher education," in Proc. 8th Int. Conf. Comput. Methodol. Commun. (ICCMC), Tamil Nadu, India, Mar. 2025, doi: 10.1109/ICCMC65190.2025.11140659.

[10] R. Waikar and S. Tamhane, "Student career guidance using generative AI," in Proc. Int. Conf. Artif. Intell. Quantum Comput.-Based Sensor Appl. (ICAIQSA), Pune, India, Dec. 2024, doi: 10.1109/ICAIQSA64000.2024.10882269.

[11] R. Sucharitha and A. Duhan, "Online career guidance and counseling - Picxida," in Proc. 9th Int. Conf. Rel., Infocom Technol. Optim. (ICRITO), Noida, India, Sep. 2021, doi: 10.1109/ICRITO51393.2021.9596186.

[12] A. Barbadekar and V. Bhagwan, "Career navigator: Your AI career counselor," in Proc. 2nd DMIHER Int. Conf. Artif. Intell. Healthcare, Educ. Ind. (IDICAIEI), Nagpur, India, Mar. 2024, doi: 10.1109/IDICAIEI61867.2024.10842724.

[13] M. Jawhar and Z. Bitar, "AI-powered customized university and career guidance," in Proc. Intermountain Eng., Technol. Comput. (IETC), Utah, USA, May 2024, doi: 10.1109/IETC61393.2024.10564423.

[14] A. Parola and L. S. Sica, "Enhancing career-related teacher support through technologies: The NEFELE training model," in Proc. IEEE Int. Conf. Metrology eXtended Reality, AI Neural Eng. (MetroXRAINE), Milan, Italy, Oct. 2023, doi: 10.1109/METROXRAINE58569.2023.10405794.

[15] Q. Wang et al., "AdaptJobRec: Enhancing conversational career recommendation through an LLM-powered agentic system," arXiv preprint arXiv:2508.13423, Aug. 2025.

[16] R. Petkov, "Digital media literacy, artificial intelligence and modernization of youth career services," in Proc. XXX Int. Sci. Conf. Electron. (ET), Sozopol, Bulgaria, Sep. 2021, doi: 10.1109/ET52713.2021.9579955.

[17] U. Bhargava and N. Koul, "CounselAI: Transforming career counseling with Gen AI for personalized multilingual guidance," in Proc. Int. Conf. Softw., Knowl., Inf. Manage. Appl. (SKIMA), Chiang Mai, Thailand, Dec. 2025, doi: 10.1109/SKIMA66621.2025.11155519.

Open Access This chapter is licensed under the terms of the Creative Commons Attribution-NonCommercial 4.0 International License (http://creativecommons.org/licenses/by-nc/4.0/), which permits any noncommercial use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license and indicate if changes were made.

The images or other third party material in this chapter are included in the chapter's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the chapter's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

